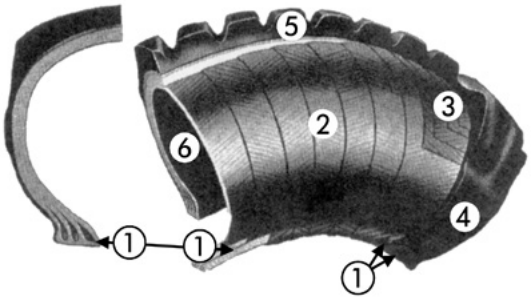
TIRE CONSTRUCTION

The pneumatic tire is essentially a flexible pressure vessel utilizing structural members (nylon, steel cable, etc.) to contain the hoop tension resulting from the inflation pressure. Rubber is utilized as a protective coating and sealant over the structural members and makes up the tread pattern which provides the wearing medium at the ground interface. The following brief explanation of the various tire constructions will assist you in selecting tires for your specific application.

Two distinct tire constructions approved on all Cat® machines are the BIAS PLY and RADIAL PLY tires. Radial tires are designated by an "R" while a "-" represents a Bias constructed tire. For example, a 45/65-45 tire would be of Bias construction and a 45/65R45 would be of Radial construction. The following is a brief explanation of the principal features of these two constructions.

Bias Ply

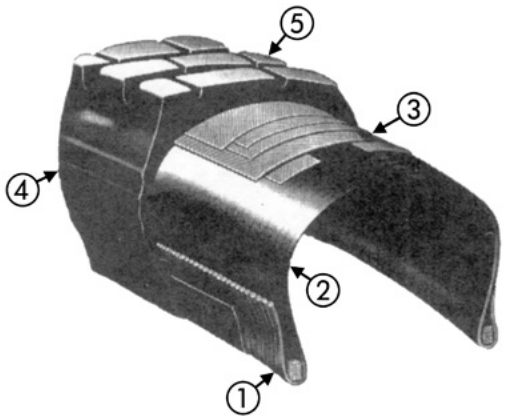
1. *Beads* — The tire beads consist of steel wirebundles (3 or 4 in larger tires) which are forced laterally by tire inflation pressure to wedge the tire firmly on the rim's tapered bead seat. The nylon plies tie into the bead bundles. The forces inherent in the tire are transmitted from the rim through the bead bundles into the nylon.
2. *Body plies* — Layers of rubber-cushioned nylon cord comprise the tire casing. Alternating plies of cord cross the tread centerline at an angle (bias). The term "ply rating" is an index of tire strength and not the actual number of tire plies.



Bias Ply Construction

3. *Breakers or tread plies* — These, if used, are confined to the tire's tread area and are intended to improve casing strength and provide additional protection to the body plies. Some "work" tires employ steel breakers or belts to further protect the casing.
4. *Sidewalls* — These are the protective rubber layers covering the body plies in the sidewall.
5. *Tread* — The wearing part of the tire that contacts the ground. It transmits the machine weight to the ground and provides traction and flotation.
6. *Inner liner* — This is the sealing medium that retains the air and, combined with the "O" ring seal and rim base, eliminates the need for inner tubes and flaps.
7. *Tubes and flaps (not shown)* — Required if the tire is not of tubeless construction with an inner liner.
8. *Undertread* — Protective rubber cushion lying between tread and body ply.

Radial Ply



Radial Ply Construction

1. *Beads* — A single bead bundle of steel cables or steel strip (spiraled like a clock spring) comprise the bead at each rim interface.
2. *Radial casing* — This consists of a single layer or ply of steel cables laid archwise (on the radian) bead to bead.
3. *Belts* — Several layers or plies of steel cable form the belts which underlie the tread area around the tire circumference. The cable in each belt crosses the tread centerline at an angle with the angle being reversed from the preceding belt.
4. *Sidewalls.*
5. *Tread.*
6. *Undertread* — Protective rubber cushion lying between tread and steel belts.

Bias and Radial Tire Advantages

	Bias	Radial
Tread Life		X
Heat Resistance		X
Cut Resistance — Tread		X
Cut Resistance — Side Wall	X	X
Traction		X
Flotation		X
Stability	X	
Fuel Economy		X
Repairability		X

TIRE TYPES

Off-the-road tires are classified by application in one of the following three categories:

1. *Transport tire* — For earthmoving machines that transport material such as trucks and wheel tractors.
2. *Work tire* — Normally applied to slow moving earthmoving machines such as graders and loaders.
3. *Load and carry* — Wheel loaders engaged in transporting as well as digging.

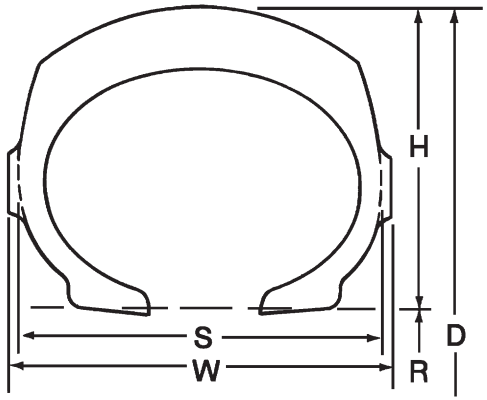
TIRE SIZE NOMENCLATURE

Tire size nomenclature is derived from the approximate cross section width and rim diameter typically in the format of Tire Width, Aspect Ratio, and Rim Diameter (example: 45/65-45). Available tire types include:

1. A wide base tire has a section height to section width ratio in the range of 0.83. As an example, a 29.5-25 tire has an approximate cross section width of 749 mm (29.5") (first number) and a rim diameter of 635 mm (25") (second number).
2. A conventional tire has a section height to section width ratio in the range of 0.96. As an example, a 24.00R35 tire has an approximate cross section width of 610 mm (24") (first number) and a rim diameter of 889 mm (35") (second number).
3. A low profile tire has a section height to section width ratio in the range of 0.65. As an example, a 45/65-45 tire has an approximate cross section width of 1143 mm (45") (first number), a 65% aspect ratio designated as 65 (second number), and a rim diameter of 1143 mm (45") (third number).

   If designated 45/65 R39, then the R denotes radial construction.

When comparing a wide base tire to a standard base tire, a larger first number on a wide base tire with the same rim diameter does not mean the wide base is larger in overall diameter. For example, the 18.00-25 conventional tire is larger in diameter than the 20.5-25 wide base tire. The 18.00-25 is comparable in overall diameter to the 23.5-25 wide base tire.



Tire cross-section

D = Tire Overall Diameter
R = Nominal Rim Diameter
H = Tire Section Height
S = Tire Section Width
W = Tire Width (includes ornamental ribs)
$\frac{H}{S}$ = Aspect Ratio

CODE IDENTIFICATION FOR OFF-HIGHWAY TIRES

The tire industry has adopted a code identification system to be used for off-the-road tires. This identification system will reduce the confusion caused by the trade names for each type of tire offered by each tire manufacturer. The industry code identification is divided into six main categories by types of service as follows:

C — Compactor Service
E — Earthmover Service
G — Grader Service
L — Loader and Dozer Service
LS — Log-Skidder Service
F — Industrial
R — Agricultural Tractor
I — Agricultural Implement

The sub-categories are designated by numerals, as follows:

Code Identification

		% Tread Depth
Compactor		
C-1	Smooth	100
C-2	Grooved	100
Earthmover		
E-1	Rib	100
E-2	Traction	100
E-3	Rock	100
E-4	Rock Deep Tread	150
E-7	Flotation	80
Grader		
G-1	Rib	100
G-2	Traction	100
G-3	Rock	100
G-4	Rock Deep Tread	150
Loader and Dozer		
L-2	Traction	100
L-3	Rock	100
L-4	Rock Deep Tread	150
L-5	Rock Extra Deep Tread	250
L-3S	Smooth	100
L-4S	Smooth Deep Tread	150
L-5S	Smooth Extra Deep Tread	250
L-5/L-5S	Half Tread Extra Deep	250
Log-Skidders		
LS-1	Regular Tread	100
LS-2	Intermediate Tread	125
LS-3	Deep Tread	150
HF-4	Extra Deep Tread	250
Industrial		
F-3	Traction Tread	
Agricultural Tractor		
R-1	Regular Tread	
R-3	Shallow Tread	
R-4	Industrial Tractor	
Agricultural Implement		
I-3	Tractor Tread	

For additional information about tire specifications, tread pattern, air pressure, TKPH, etc, please refer to the Manufacturer Website:

Michelin	MichelinEarthMover.com
Bridgestone	Bridgestone.com
Maxam	MaxamTire.com
Camso	Camso.co
Triange	TriangleTiresUS.com
Firestone	FirestoneTire.com
Titan	Titan-Intl.com
Continental	Continential-Tires.com
Yokohama	Y-Yokohama.com
Goodyear	GoodyearOTR.com

RADIAL TIRE IDENTIFICATION

Code Identification for Michelin Tires

All Michelin earthmover tires are radial construction, designated by the "X" marking. They contain a single steel radial ply with a series of steel belts placed around the tire's circumference which reinforce and stabilize the tread.

Following are the tread designs currently available from Michelin with the different internal constructions depending on the application.

Type A4 Particularly resistant to cuts, tread tearing and abrasion on very rough surfaces.

Type A Particularly resistant to cuts, tread tearing and abrasion at average speeds which are higher than those for A4 (above).

Type MB4 Compromise solution between abrasion resistance and average speed on rough surfaces (from 49 inches) with a higher wear resistance than the Type B4.

Type B4 Compromise solution between abrasion resistance and average speed on rough surfaces.

Type B Higher resistance to internal heat generation on surfaces which are not particularly rough.

Type MC4 Adapted to running on long cycles at high speeds on well-maintained roads with a higher wear resistance than the Type C4.

Type C4 Adapted to running on long cycles at high speeds on well-maintained roads.

Type C Very high resistance to high/average speeds on long cycles running on well-maintained roads.

Since Michelin radial tires contain a single steel casing ply, they utilize the industry method of designating radial tire strength in terms of "stars." Their system consists of a one star, two star, and three star rating as an indication of the tire's carrying capacity. The one star is the lightest construction, generally used on work and slow moving transport machines. Two star tires are used on most medium and high speed transport machines. Three star construction provides the greatest carrying capacity for a given size and is only available in small standard base tires.

This combination of tread designs and types of construction provides a range of radial tires which cover most earthmoving applications. We recommend that in applying steel radial tires to your machines you provide all site condition data to the tire manufacturer. Obtain their recommendations as to which tire will provide the most economical operation.

Code Identification for Goodyear Radial Tires

All Goodyear steel radial earthmover tires have been designated *Unisteel* followed by a three or four digit alpha-numeric code that identifies the particular tread. For example, for a RL-2+, RL stands for Rock Lug and indicates that the upper sidewall has rock protection. The number in the code corresponds to the tire industry identification system (2-traction, 3-rock, etc). The fourth digit, if any, is used to designate tread design differences for the same basic tread type.

Following are the radial tread designs currently available from Goodyear with the compound and construction types depending on the application.

Compound Description	Compound Code
High Heat Resistant	2
Heat Resistant	3
Standard Abrasion Resistant	4
Ultra Abrasion Resistant	6

Construction Description	Construction Code
Standard	S
Heavy Duty	H
Extra Heavy Duty	HW
Steel Breakers	J
Heavy Undertread	U
Low Angle Belts	SL

	Tread Compounds			
Tread Design	2S	4S	6S	Primary TRA Code(s)
AT-2A	X	X	X	E-3/L-3
GP-2B	X	X	X	E-3/G-3/L-3
GP-3D	X	X	X	E-3/L-3
GP-4B	X	X	X	E-4
GP-4B AT	X	X	X	E-4/G-4
GP-4C	X	X	X	E-4/G-4/L-4
GP-4D	X	X	X	E-4/L-4
RL-2+	X	X	X	E-3/G-3/L-3
RL-2F	X	X	X	E-2/G-2/L-2
RL-3+	X	X	X	E-3
RL-3A	X	X	X	E-3
RL-3J	X	X		E-3
RL-4	X	X	X	E-4
RL-4A	X	X	X	E-4
RL-4B	X	X	X	E-4
RL-4H	X	X	X	E-4
RL-4H II	X	X	X	E-4
RL-4J	X	X	X	E-4
RL-4J II	X	X	X	E-4
RL-4K		X	X	L-4
RL-4M+	X	X	X	E-4
RL-5K		X	X	L-5
RM-4A+	X	X	X	E-4
RM-4B+				
RT-3A+	X	X	X	E-3
RT-3B			X	L-3
RT-4A	X	X	X	E-4
RT-4A+	X	X	X	E-4
TL-3A+	X	X	X	E-3/L-3

A star rating system instead of the ply rating system indicates the casing strength of radial tires. These symbols indicate the recommended inflation for a particular tire load. Following the star rating code is Goodyear's Custom Compound and Construction code. For a tire designated "2S" the 2 indicates a heat resistant compound and the S indicates standard construction. The higher the number the greater the abrasion and cut resistance with a corresponding lower TKPH/TMPH rating.

Code Identification for Bridgestone Radial Tires

The Bridgestone steel radial earthmover has been designated as V-Steel. Following are the radial tread designs currently available from Bridgestone with the compound and construction types depending on the application.

Bridgestone Compound and Structure Codes

1A	Standard
2A	Cut-Resistant
2V	Special Cut Resistant (Steel Breaker)
2Z	Special Cut Resistant (Side Steel Breaker)
3A	Heat-Resistant
E	Earthmover
G	Grader
D	Loader and Dozer
S	Logging

Tread Design	Tread Name	Tread Compounds 1A	2A	2V	3A	Primary TRA Code(s)
VELS	V-Steel E-Lug S	X	X		X	E-4
VELSL	V-Steel E-Lug S				X	E-4
VFT	V-Steel F-Traction	X	X		X	E-2
VHS	V-Steel H-Service				X	E-2
VJT	V-Steel J-Traction		X			E-3/L-3
VKT	V-Steel K-Traction	X	X			E-2/G-2/L-2
VLT	V-Steel L-Traction	X	X		X	E-3/L-3
VLTS	V-Steel L-Traction S		X			E-4/L-4
VMT	V-Steel M-Traction	X	X		X	E-3/L-3
VMTP	V-Steel M-Traction Premium	X	X		X	E-4
VMTS	V-Steel M-Traction S	X	X		X	E-4/G-4
VRDP	V-Steel Rock Deep Premium	X	X		X	E-4
VREP	V-Steel Rock E-Premium	X	X		X	E-4
VRF	V-Steel Rock Fast	X			X	E-3
VRL	V-Steel R-Lug	X				E-3
VRLS	V-Steel R-Lug S	X	X		X	E-4
VRPS	V-Steel Rock Premium Service	X	X		X	E-4
VRQP	V-Steel Rock Quarry Premium		X			E-4
VSB	V-Steel S-Block		X			E-2
VSDL	V-Steel D-Lug		X			L-5
VSDT	V-Steel Super Deep Traction		X			L-5
VSMS	V-Steel Smooth Tread-MS		X	X		L-5S
VSNL	V-Steel N-Lug		X			L-4
VSNT	V-Steel N-Traction		X			E-4/L-4
VSW	V-Steel Snow Wedge		X			G-2/L-2
VTS	V-Steel Traction-Stability		X			L-3
VUT	V-Steel Ultra Traction		X			G-2/L-2
VZTP	V-Steel Z-Traction Premium	X	X		X	E-4
VZTS	V-Steel Z-Traction S	X	X		X	E-4

The casing strength, i.e., load carrying capacity of tire is indicated by star rating system; 1-star, 2-star and 3-star. Bridgestone's Off-the-Road tires are designed and produced to meet the commonly accepted international standards, those set by the TRA (Tire and Rim Association) in the U.S.A., by the ETRTO (European Tire and Rim Technical Organization) in Europe and/or by the JATMA (Japan Automobile Tire Manufacturers' Association) in Japan. Where differences exist between the TRA, ETRTO and JATMA standards, Bridgestone selects the most appropriate.

Code Identification for Eurotire Radial and Bias Tires

Eurotire manufactures a range of bias-ply and radial tires for use on mining equipment. Eurotire steel radial tires are designated with an "EU." The following are the radial tread designs currently available from Eurotire with the compound and construction types for a variety of applications.

Compound Description	Compound Code
Heat Resistant	H
Intermediate	S
Abrasion Resistant	A

	Compound Code			
Tread Design	H	S	A	TRA Code(s)
EUROK	X	X	X	E4
EUTRAK	—	X	X	E3, E4, L4

Eurotire radial tires are marked with a star rating system to indicate casing strength. Eurotire manufactures 2 star radial tires.

It is important to understand the specific usage and operating conditions of your job site so that the most appropriate tire choice can be determined. Eurotire representatives can help you make the most informed decision regarding tire choice, including weighing the benefits of radial vs. bias, selecting between different tread designs and compounds and providing service solutions to best suit your specific needs.

TON-KILOMETER PER HOUR (TKPH)

Tire selection and machine operating practices have, in some cases, become the critical factors in the over-all success of earthmoving ventures. One of the most serious problems occur when tires are operated at temperatures above their capabilities. Separation and related failures occur. To help you avoid temperature related failures, Caterpillar has been instrumental in developing the *Ton-Kilometer Per Hour* (TKPH), also known as *Ton-Mile Per Hour* (TMPH), method of rating tires. The formula to convert a TKPH rating to a TMPH rating is:

$$TMPH = TKPH \times 0.685$$

Heat and Tire Failure

Tire manufacturing requires heat in the vulcanizing process converting crude rubber and additives into a homogeneous compound. The heat required is typically above 132° C (270° F).

A tire also generates heat as it rolls and flexes. Heat generated faster than it can be radiated into the atmosphere gradually builds within the tire and reaches maximum level at the outermost ply or belt.

Over time, enough heat can develop from overflexing to actually reverse the vulcanizing process or "revert" the rubber causing ply separation and tire failure. Only a brief time at reversion temperature initiates the failure. Experience shows that few pure heat separation cases occur. Most so-called heat separations are in tires operating below the reversion level.

As a tire's operating temperature increases the rubber and textiles within significantly lose strength. The tire becomes more susceptible to failures from cornering, braking, impact, cut through, fatigue and heat separation. If operating tires at higher temperatures is absolutely necessary, it is essential the machines be operated to reduce the probability of premature tire failure. No hard cornering without superelevation, no panic braking, etc.

The TKPH formula was developed to predict tire temperature buildup. The system is a method of rating tires in proportion to the amount of work they can do from a temperature standpoint. It utilizes the product of *load* × *speed* to derive an index of the tire temperature buildup. Even at or below a tire's TKPH, failures may be initiated by overstressing the tires.

It is possible by using a needle type pyrometer to measure temperature at any desired point within the tire casing. However, the instrumentation and the technique does not lend itself to general field use. The greatest difficulty is locating the thickest (therefore the hottest) tread bar in any given tire using giant calipers. The tire must then be drilled along the centerline of this bar from shoulder to shoulder at 52 mm (2") intervals. These 3.18 mm (1/8") diameter holes extend down through the tread and undertread to the topmost reinforcement. This procedure is fully described under SAE Recommended practice J1015.

The TKPH rating system as given in this SAE specification is approved by most tire manufacturers. Michelin, in addition to providing TKPH ratings has developed their own speed/load carrying rating system and we recommend that Michelin be consulted where high tire temperatures are a concern.

Heat generation in a specific tire at recommended pressure depends on three factors:

- the weight the tire is carrying (flex per revolution),
- the speed the tire is traveling over the ground (flexures over a period of time), and
- the air temperature surrounding the tire (ambient temperature) and road surface temperature.

Once a tire manufacturer has determined a tire's temperature characteristics and expressed them in TKPH, the above listed specific job conditions can be used to determine any tire's maximum work capacity. These conditions provide on site ability to predict and avoid costly tire separations.

Ton-Kilometer-Per-Hour Rating System

The tire TKPH can be matched to the site TKPH as well as compared with TKPH values of different makes and types of tires.

TKPH Job Rate

Average Tire Load × Average Speed for the shift

Average Tire Load

$$\frac{\text{“Empty” tire load + “loaded” tire load}}{2}$$

Average Speed

$$\frac{\text{Round trip distance in kilometers} \times \text{number of trips}}{\text{Total Hours (in the shift)}}$$

For excessive haul length (32 kilometers or more) consult your tire representative for modification to the TKPH value.

To use in the United States Customary System, change kilometers to miles and use short tons.

It should be noted that prolonged operation at high casing temperatures can fatigue the nylon at the flex points in the sidewalls.

The following are the most recent TKPH ratings as made available by Goodyear, Michelin and Bridgestone, and are subject to change on their part at any time. Other tire manufacturers' TKPH ratings will be included in future handbook editions when and if made available. For latest TKPH ratings, consult specific tire manufacturer at time of machine and/or tire purchase.

Load-and-Carry TKPH

The wheel loader, when used in load-and-carry applications, may encounter temperature problems similar to those normally associated only with tires on scrapers, trucks and wagons. **Do not place the vehicle in load-and-carry applications without first consulting the tire manufacturer, or obtaining maximum load and speed ratings and pressure recommendations from the tire manufacturer.**

Conventional and Radial Steel Cord Tire Options

Tire options now provide types to operate in conditions ranging from rock and abrasive materials, to jobs with high speed hauls in good materials.

The best tire type can be different for the drive tires than for other tires on the same machine. TKPH should be calculated for all tires.

TIRE DRIVE-AWAY RECOMMENDATIONS

Heat separation can be a problem during machine delivery and moving machines from one job to another. Whenever roading earthmoving machines, *check your supplier for the tire manufacturer's recommended speed limitations on the specific tires involved.*

Some tire manufacturers also recommend that vehicles equipped with extra tread depth or special compounded tires should not be roaded without their specific approval. Our tests support this recommendation, especially for L-3, L-4, E-4 and L-5 tires.

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Because of the variance between specific tires it is recommended that at the time of purchase you check with your tire supplier for the manufacturer's specific TKPH ratings for the tires purchased.

TIRE AND RIM ASSOCIATION RATINGS

While the TKPH Rating System provides a method to determine the tire's work capacity, Tire and Rim Association Ratings provide a guide for evaluating a tire's structural capacity. These two rating systems should be used in conjunction to evaluate tire performance.

TIRE SELECTION

Selecting the optimum tire for a given application is particularly critical for earthmoving. The machines have the capability to outperform the tires and, unless proper practices are observed, very costly premature tire failures can occur. Job conditions vary greatly throughout the world, as well as within any given job site, and selecting the optimum tire requires careful consideration of all factors involved. In general, the tire manufacturer should be consulted before making the selection for any given application. In some cases, the tire manufacturer can fabricate tires specifically tailored for a given job site.

For those applications where wear is extremely slow, especially as a result of only occasional operation throughout the year, the cheapest lightweight tire needs to be given strong consideration.

As job conditions become severe, the following factors should be evaluated in selecting a tire:

Transport or Load-and-carry —

- TKPH (primary consideration)
- Minimum approved star/ply rating or greater
- Largest optional size
- Thickest tread commensurate with TKPH
- Most cut resistant tread commensurate with TKPH
- Belted construction

Grader —

- Tire load rating suitable for maximum equipped machine weight (See Tire Load Worksheet on next page)
- Application specific tire (snow, construction, road maintenance, mining, general purpose, all season)
- Bias or radial based on initial cost, puncture resistance, rolling resistance, life to retread/repair

Loader or Dozer —

- Minimum approved ply rating or greater
- Largest optional size
- Thickest tread
- Thickest available undertread
- Buttressed shoulder
- Most cut resistant tread
- Belted construction
- Lowest aspect ratio

All tires should be operated at the tire manufacturer's recommended inflation pressure for a given application. Inflation pressure should be checked every working day with an accurate gauge. This gauge should be checked against a known standard such as a dead weight tester at least once a month.

Excess loads may result from factors such as varying material density, field modifications to equipment, mud accumulation, load transfer, etc. Only under these conditions may the actual in service tire load exceed the rated machine load. When excess loads are encountered, cold inflation pressures **must** be increased to compensate for higher loads. Increase tire inflation pressure 2% for each 1% increase in load.

	Maximum Excess Load	Pressure
Bias Ply	15%	30%
Radial Ply	7%	14%

The above loads will result in reduced tire performance and must be approved by the tire manufacturer.

The use of chains is difficult to justify except under a few conditions. Chains are very costly and heavy, and require more maintenance than most operations can provide. On some models sufficient clearance does not exist for chains with all tire combinations. Extensive modifications may be required if chains are needed for the job.

Foam filling tires is normally not recommended due to high cost and lack of local filling facilities. Its use should be confined to loader and dozer applications where penetrations occur almost daily. If foam is used be sure to adhere to recommended equivalent pressures of nitrogen and use highest available ply rating. Consult tire manufacturer for specific warranty concerns.

TIRE SELECTION GUIDE

Material	Road or Ground Condition	Treads		
		Haul Trucks and Wheel Tractor-Scrapers	Wheel Tractors or Wheel Loaders	Graders*
Silt and clay: – No rock – High moisture content	– Good varying to poor – High rolling resistance	E-2 (Traction)	L-2 (Traction)	G-2 (Traction)
Silt and clay: – Some rock – Variable moisture content	– Good varying to poor	E-3 (Rock) E-2 (Traction)	L-3/L-4/L-5 (Rock) L-2 (Traction)	G-3/G-4 (Rock) G-2 (Traction)
Silt/clay/gravel/sand: – Low moisture content	– Excellent to good – Firm surface	E-3/E-4 (Rock)	L-3/L-4/L-5 (Rock)	G-3/G-4 (Rock) L-3/L-4/L-5 (Rock)
Silt/clay/gravel/sand: – High moisture content	– Poor – Rutted – Pot holes	E-3/E-4 (Rock)	L-3/L-4/L-5 (Rock)	G-3/G-4 (Rock) L-3/L-4/L-5 (Rock)
Blasted rock	Hard surface, rough	E-4 (Rock)	L-5 (Rock) L-5S (Smooth)	G4 (Rock) L-4/L-5 (Rock)
Sand: – Very low silt/clay content	Good to fair surface	E-3 (Rock) E-7 (Flotation)	L-3 (Rock) L-3S (Smooth)	G-3 (Rock) Low pressure

*__NOTE:__ In some cases, an L type tire is appropriate for use on a Grader application, consult your tire supplier for proper tire selection.

Optimal pressures may vary depending on specific applications and working conditions. Always consult your local tire supplier for operating pressures.

	BIAS PLY TIRES						RADIAL PLY TIRES					
	Weight Increase Per Tire		Mixing Proportions				Weight Increase Per Tire		Mixing Proportions			
			CaCl		Water				CaCl		Water	
Tire Size	kg	lb	kg	lb	L	gal	kg	lb	kg	lb	L	gal
13.00 × 24TG	188	**414**	55	**122**	132	**35**	185	**407**	57	**125**	128	**34**
14.00 × 24TG	215	**475**	63	**140**	151	**40**	256	**565**	79	**173**	179	**47**
15.5 × 25	192	**423**	56	**125**	136	**36**	224	**493**	69	**151**	155	**41**
16.00 × 24TG	333	**735**	98	**217**	234	**62**	355	**783**	109	**240**	246	**65**
17.5 × 25	262	**577**	77	**170**	185	**49**	311	**686**	95	**210**	216	**57**
18.00 × 25	454	**1002**	134	**296**	322	**85**	502	**1107**	154	**340**	348	**92**
18.4 × 34	417	**919**	123	**272**	295	**78**	—	**—**	—	**—**	—	**—**
20.5 × 25	405	**892**	119	**263**	284	**75**	448	**987**	137	**303**	310	**82**
23.1 × 26	522	**1151**	154	**340**	367	**97**	—	**—**	—	**—**	—	**—**
23.5 × 25	585	**1291**	173	**382**	412	**109**	633	**1396**	194	**428**	439	**116**
24.5 × 32	703	**1549**	207	**458**	496	**131**	—	**—**	—	**—**	—	**—**
26.5 × 25	758	**1671**	224	**494**	533	**141**	841	**1853**	258	**568**	583	**154**
26.5 × 29	752	**1658**	222	**490**	530	**140**	928	**2045**	284	**627**	644	**170**
28L × 26	709	**1563**	209	**462**	500	**132**	—	**—**	—	**—**	—	**—**
29.5 × 25	970	**2139**	286	**632**	685	**181**	1073	**2368**	328	**723**	745	**197**
29.5 × 29	1050	**2315**	310	**684**	738	**195**	1190	**2623**	365	**804**	825	**218**
875/65 × 29	—	**—**	—	**—**	—	**—**	1445	**3186**	429	**946**	1016	**268**
29.5 × 35	1159	**2556**	344	**758**	821	**217**	1286	**2835**	394	**869**	892	**236**
30.5L × 32	874	**1928**	258	**570**	617	**163**	—	**—**	—	**—**	—	**—**
33.25 × 35	1485	**3275**	439	**968**	1048	**277**	1592	**3508**	487	**1074**	1105	**292**
37.25 × 35	1712	**3775**	505	**1115**	1211	**320**	2128	**4692**	653	**1439**	1476	**390**
38 × 39	1870	**4123**	552	**1218**	1317	**348**	—	**—**	—	**—**	—	**—**
35/65 × 33	1339	**2953**	396	**873**	942	**249**	1430	**3152**	438	**967**	992	**262**
40/65 × 39	2077	**4580**	614	**1353**	1465	**387**	2194	**4836**	673	**1483**	1522	**402**
41.25/70 × 39	1897	**4183**	561	**1236**	1336	**353**	—	**—**	—	**—**	—	**—**
45/65 × 45	2548	**5617**	753	**1659**	1794	**474**	—	**—**	—	**—**	—	**—**

NOTE: Ballast weight for bias ply tires from Goodyear data, radial ply weights from Michelin data. Contact your tire supplier for additional information. Under abnormal tire wear conditions, ballasting of rear tires may be desirable. Ballasting of front tires also should only be done where extremely rapid tire wear rates are encountered. Excessive weight will reduce machine performance.

NOTE: Fillage beyond 75% of tire enclosed volume is not recommended. With liquid ballasting, inflation pressure must be checked at least once per day.

NOTE: 1.6 kg **(3½ lb)** Calcium Chloride per gallon water. Solution weighs 4.6 kg **(10.15 lb)** per gallon.

NOTE: Total machine mass including all attachments in operating condition, all reservoirs at full capacity and ballasted tires must not exceed certification mass listed on the ROPS certification label.

NOTE: Special air to water valves are required for liquid filled tires.

Consult the tire manufacturer before adding Ballast to any tire.

SEBD0351-50

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